



Security Culture 2.0

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Frontispiece

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SECUR-E Developer Security Self-Assessment Tool

Deliverable 3 - Scoring & Persona Classification Framework

A complete six-step scoring protocol mapping COM-B subscale scores to SECUR-E personas, with decision rules for borderline profiles, a confidence rating system, and fully worked examples. Grounded in [DASP](#) (Larios-Vargas et al., 2023)¹, the [COM-B Model](#) (Michie et al., 2011)² and the SECUR-E Personas Framework.

Overview: How Classification Works

The scoring system works in six sequential steps. The first two are mechanical: calculating each subscale's raw total and converting it to a normalized score on a 0-100 scale. The next two are interpretive: placing each normalized score into a behavioral band and then comparing the resulting profile against the six SECUR-E persona patterns. The final two handle the messiness of real data: a set of explicit decision rules for the five most common borderline cases and a confidence rating that indicates how certain any given classification should be.

A critical design principle runs through all six steps: the classification unit is always the three-dimensional profile, the combination of *Capability* (CAP), *Opportunity* (OPP), and *Motivation* (MOT) scores, not any single subscale number. SECUR-E is explicit that personas are 'coherent combinations of characteristics,' not positions on a single scale. Two developers can have identical MOT scores but completely different personas depending on their CAP and OPP scores.

A composite single-number score is also calculated and documented in Step 6, but it serves only as a summary communication tool for reporting and tracking; it is never used as the basis for persona classification. That task always requires the full three-dimensional profile.

Step 1: Calculate Raw Subscale Scores

Apply reverse scoring, then sum each section independently.

For each of the three subscales, complete the following procedure before doing anything else. The order matters; reverse-scoring must happen before summing, or the total will be wrong.

Reverse-scoring formula: Scored value = 6 minus the raw response

Apply this transformation to the following items before summing:

Section B (Capability): Items 4, 6, and 8

Section C (Opportunity): Items 12, 14, and 16

Section D (Motivation): Items 19, 21, and 23

Example: If a respondent rated item 4 as 2 (Disagree), the score is $6 - 2 = 4$. If they rated item 19 as 5 (Strongly Agree), the score is $6 - 5 = 1$.

After applying the formula to those items, sum all eight scored values within each section. The result is the raw subscale score.

¹ Larios-Vargas, E., Elazhary, O., Yousefi, S., Lowlind, D., Vliek, M. L. W., & Storey, M.-A. (2023). DASP: A Framework for Driving the Adoption of Software Security Practices. *IEEE Transactions on Software Engineering*, 49(4), 2892-2919.

² Michie, S., van Stralen, M. M., & West, R. (2011). The behaviour change wheel: A new method for characterising and designing behaviour change interventions. *Implementation Science*, 6(1), 42.

Valid raw score range per subscale: Minimum 8 (all items scored 1), Maximum 40 (all items scored 5). A score outside this range indicates a calculation error.

Step 2: Normalize to a 0-100 Scale

Convert each raw score using the formula below, then assign a behavioral band.

Normalizing to a 0-100 scale makes the three subscales directly comparable regardless of item count. It makes the band thresholds interpretable as percentages of the possible range rather than as arbitrary raw numbers. The formula is straightforward:

Normalization formula: Normalized score = $(\text{Raw score} - 8) \div 32 \times 100$

The constants 8 and 32 are the minimum raw score and the total range ($40 - 8 = 32$), respectively.

Example: Raw score of 27 $\rightarrow (27 - 8) \div 32 \times 100 = 19 \div 32 \times 100 = 59.4 \rightarrow$ rounded to 59.

Example: Raw score of 8 $\rightarrow (8 - 8) \div 32 \times 100 = 0$.

Example: Raw score of 40 $\rightarrow (40 - 8) \div 32 \times 100 = 100$.

Once normalized, each score is assigned to one of five behavioral bands. These bands are the primary language of the scoring system; they are what practitioners report and compare, rather than precise numerical values. The band boundaries correspond directly to the four-level COM-B coding scheme used in SECUR-E (High, Moderate, Low, Very Low/Absent), with an additional Very High band at the upper extreme to distinguish developers who score uniformly 4-5 across all items.

Band	Raw Score (8-40)	Normalized (0-100)	Avg. Item Response	Interpretation
Very Low	8 - 15	0 - 21	< 2.0	Security capability/opportunity/motivation is essentially absent from this dimension. Primary intervention target.
Low	16 - 22	22 - 43	2.0 - 2.8	Aware of the dimension, but engagement is inconsistent or minimal. Support needed.
Moderate	23 - 29	44 - 65	2.9 - 3.6	Adequate but not consistently strong. Responsive to opportunity or nudges.
High	30 - 36	66 - 87	3.7 - 4.5	Well-developed. Likely a strength to leverage in intervention design.
Very High	37 - 40	88 - 100	> 4.5	Exceptional. Consider as a resource or model for peers.

Step 3: Compare Profile to Persona Reference Patterns

Match the respondent's three-band profile against the six SECUR-E archetype signatures.

With band labels assigned to CAP, OPP, and MOT, the three-band combination forms a profile that can be compared against the six SECUR-E persona archetypes. The reference table below shows the expected band for each dimension in each persona, along with the approximate raw score range and the defining behavioral pattern.

Persona	CAP Band	OPP Band	MOT Band	Defining Pattern
Skeptic	Moderate 22-30	Low-Moderate 20-28	Low-Moderate 17-24	Adequate skills but genuine doubt about whether the security effort is worthwhile. ROI skepticism, not ignorance or malice. OPP is neutral-to-positive; the organization provides support, but the developer does not value it enough to act beyond the minimum. The distinguishing feature is MOT: low but not absent. Open to evidence.
Enthusiast	High-Very High 30-40	Very Low-Moderate 8-27	High-Very High 30-40	Strong technical and self-efficacy capabilities, combined with high intrinsic motivation, make security personally meaningful and intellectually engaging. Critically, OPP is often limited: they may work in organizations without security champions, dedicated time, or policy support, yet still act. Burnout risk.
Compliant	Moderate 22-29	Moderate-Very High 29-40	Very Low-Moderate 8-24	The organization does the work that motivation does not. High OPP (policies, tools, mandates, security teams) is the primary driver of behavior. CAP is adequate but capped below High; a Very High-CAP profile with this OPP/MOT pattern routes to the Compliant→Embedded transitional case instead (Step 5). Security behavior is tightly coupled to enforcement:
Unaware	Very Low-Low 8-18	<i>Not discriminating, any score 8-40; Rule 1 classifies as Unaware regardless of OPP</i>	Very Low-Low 8-18	Knowledge gaps are so fundamental that the developer cannot recognize the problem space. Low OPP may reflect an organization that provides resources the developer cannot yet use productively. MOT is very low, not because of resistance but because of invisibility; security is not on the radar. Highest urgency for capability intervention.
Resistant	Moderate 22-30	Moderate 22-28	Very Low 8-16	Capable and working in a reasonably supportive environment, but actively disengaged from or opposed to security requirements. The motivational barrier arises from psychological reactance: security constraints are perceived as threats to autonomy. More negative on MOT items about imposed restrictions than Skeptics, and less open to evidence-based persuasion.
Embedded	High-Very High 30-40	Moderate-Very High 28-40	High-Very High 30-40	Security is fully integrated into professional identity and daily practice, habitual, automatic, and intrinsically valued. All three dimensions score high. The distinguishing feature from Enthusiast is OPP: Embedded developers work in organizations that support their security orientation, creating a self-reinforcing cycle. Best placed to serve as champions or mentors.

Note on the Enthusiast's OPP profile: The Enthusiast is defined by operating in a low-to-moderate opportunity environment while maintaining high CAP and MOT. This means that an Enthusiast in an organization with a strong security infrastructure (OPP High) will not be classified as an Enthusiast; they will be classified as Embedded if MOT and CAP are also high. The Enthusiast designation specifically captures the high-motivation-despite-low-support pattern.

Step 4: Apply Classification Decision Rules

Work through the priority-ordered rules in sequence. Stop at the first rule that applies.

The six decision rules below are applied in order of priority. The priority sequence is designed to ensure that the most unambiguous, high-stakes profiles, Unaware and Embedded, are captured first, because misclassifying these two personas has the greatest cost for intervention design. Work through the rules from Rule 1 downward, stopping as soon as one applies. **If no single rule applies cleanly, proceed to Step 5 (borderline rules).**

Priority	Condition (after reverse-scoring)	Classification	Rationale
1 (Highest)	CAP ≤ 18 AND MOT ≤ 18	Unaware	Very low capability and very low motivation together constitute the defining Unaware profile. Classify regardless of the OPP score; a high OPP cannot compensate for a lack of foundational knowledge to engage with available resources. SECUR-E: 'Even when sufficient opportunities exist, low capability and low motivation combined hinder security behavior.'
2	CAP ≥ 30 AND OPP ≥ 28 AND MOT ≥ 30	Embedded	All three dimensions are High. Security is habitual, organizationally supported, and intrinsically valued. SECUR-E: 'security behavior is self-sustaining.' Classify before other high-score profiles to prevent misclassification as an Enthusiast.
3	CAP ≥ 30 AND MOT ≥ 30 AND OPP ≤ 27	Enthusiast	High CAP and high MOT with limited OPP. The Enthusiast is defined precisely by this asymmetry: intrinsic motivation compensates for an unsupportive environment. SECUR-E: 'Enthusiasts overcome limited opportunity through strong motivation.'
4	OPP ≥ 29 AND MOT ≤ 24 AND CAP 22-29	Compliant	High organizational opportunity drives behavior in the absence of strong intrinsic motivation. CAP must be at least Moderate but below High; a developer with Very High CAP who still shows this OPP/MOT pattern is better captured by the Compliant→Embedded transitional case (Step 5) than by plain Compliant. SECUR-E: 'their security behavior is coupled to organizational enforcement: strong when mandated and monitored, weak when optional.'
5	MOT ≤ 16 AND CAP 22-30 AND OPP 22-28	Resistant	Very low MOT with moderate CAP and OPP is the Resistant signature. Distinguish from Compliant (lower OPP) and Skeptic (MOT is less extreme). Resistant developers understand security requirements but actively resent them. SECUR-E: 'psychological reactance; they view security processes as adversaries to productivity.' Check the responses to MOT-3 and MOT-19 as confirmatory signals.
6	CAP 22-30 AND OPP 20-28 AND MOT 17-24	Skeptic	Moderate across all three dimensions, with MOT in the low-to-moderate range. The Skeptic is a 'comfortable middle' profile: capable enough to do security work, operating in a reasonably supportive

Priority	Condition (after reverse-scoring)	Classification	Rationale
			environment, but not convinced of its value. DASP D12 and D15 are the key drivers; the developer lacks conviction about security ROI. Lower MOT than Resistant, but more open to persuasion.

Rule 2a (Refinement, applied after Rule 2): Embedded - Exemplar

This is the one place in Step 4 where the Very High band (37-40) changes an outcome rather than serving only as a reporting nuance elsewhere in the instrument. It is not a seventh priority rule competing for the "first match wins" sequence; it is an extension checked only after Rule 2 has already fired.

Condition	Classification	Rationale
Rule 2 has fired (Embedded), AND CAP ≥ 37 AND OPP ≥ 37 AND MOT ≥ 37	Embedded (Exemplar)	All three dimensions are not merely High but Very High, uniformly near-ceiling responses. Step 2 already describes the Very High band as 'exceptional... consider as a resource or model for peers,' but until this refinement nothing in the classification logic acted on that description. This flag exists to surface the strongest candidates for formal champion or mentor roles (D6), not to introduce a new persona: an Embedded (Exemplar) profile is still Embedded for every other purpose in the instrument (D4 team grids, D5 triggers, D6 base intervention guidance). Every other rule in Step 4 continues to treat High and Very High identically; this is the sole exception.

If Rule 2 fires but any of the three dimensions falls in High (30-36) rather than Very High (37-40), report plain **Embedded** with no exemplar flag.

Important: profiles that fall between rules

In practice, approximately 20-30% of profiles will not be cleanly captured by a single rule. This is expected; SECUR-E explicitly acknowledges that 'real developers exhibit mixtures of persona characteristics.' When a profile appears to satisfy parts of two or more rules, or falls in a boundary zone between them, proceed directly to Step 5.

Do not attempt to force a classification by selecting the 'closest' rule. A dual classification or a borderline flag is a more accurate and more useful result than a false certainty.

Step 5: Apply Borderline Decision Rules

Five specific ambiguous cases, each with discriminating signals and a decision rule.

The five borderline cases outlined below highlight the most empirically grounded ambiguities within the SECUR-E persona framework. These cases have been identified from various sources: the dual classification of two participants, one categorized as Skeptic/Compliant and the other transitioning from Compliant to Embedded, as observed in the DASP study. Additionally, theoretical distinctions have been made between Skeptic and Resistant personas, and similarities in structure between Unaware and Resistant personas have been

noted in their motivational dimensions. Each case specifies the key distinguishing signal that is most likely to clarify the ambiguity without requiring further data collection.

Ambiguous Pair	Trigger Condition	Discriminating Signal	Decision Rule
Skeptic vs. Resistant	<i>Both have MOT 14-22 and CAP/OPP 22-30</i>	MOT-3 (item 19) and MOT-7 (item 23) responses. Resistant typically rates both 4-5 (strong agreement with reactance statements). Also check Reflection 2: a Resistant will rarely volunteer a security initiative; a Skeptic may, if they find evidence of value.	If $MOT \leq 16$ AND item 19 ≥ 4 : classify Resistant. If MOT 17–24: classify Skeptic with Resistant risk flag.
Enthusiast vs. Embedded	<i>Both CAP ≥ 30 and MOT ≥ 30</i>	OPP score is the primary discriminator. Also check OPP-3 (item 11: access to security champion) and OPP-5 (item 13: dedicated time). Enthusiasts in low-OPP environments often score highly on MOT-8 (item 24: self-directed learning) as a compensatory signal.	If $OPP \geq 28$: classify Embedded. If $OPP \leq 27$: classify Enthusiast. Check Reflection 3. Enthusiasts often name specific organizational deficits.
Compliant vs. Skeptic	<i>Both have CAP 22-30 and MOT 16-25</i>	OPP is the tiebreaker. Compliant has genuinely high organizational support; Skeptic has moderate OPP but doesn't leverage it due to motivational doubt. Also check MOT-5 (item 21: follows rules because required): Compliant rates this 4-5; Skeptic often 2-3 (not fully rule-driven either).	If $OPP \geq 29$: classify as Compliant. If $OPP 20-28$: classify Skeptic. Dual-code as Skeptic/Compliant if MOT-5 (item 21) ≥ 4 AND OPP 25-30.
Unaware vs. Resistant	<i>Both have very low MOT (≤ 18)</i>	CAP is the discriminator. Unaware: CAP ≤ 18 . Resistant: CAP 22–30. Also, qualitatively: Resistant will express strong opinions about security (even negative ones) in Reflection 1; Unaware will often report having no particular views or experiences.	If $CAP \leq 18$: Unaware regardless of MOT framing. If CAP 19-22 (boundary zone), apply confidence level Low; note as Unaware-Resistant risk, or vice versa, depending on MOT item content.
Compliant → Embedded (transitional)	<i>OPP ≥ 28 AND MOT 17-30 AND CAP 26-34</i>	A participant exemplifies this. The emerging Embedded profile shows both compliance AND improvement motivation. Check Reflection 2: A transitional developer will be able to name at least one unprompted example of a security initiative.	Classify as Compliant with the Embedded pathway flag. Recommend autonomy-building interventions (DASP D8, D15) to shift from extrinsic to intrinsic motivation as the primary intervention goal.

For profiles that remain unresolved after consulting this table, apply the Low confidence rating (Step 6) and report the two most plausible personas. The open reflection responses from Section E of the survey are always available as qualitative evidence to supplement quantitative ambiguity.

Step 6: Calculate Composite Score and Assign Confidence Rating

One summary score for tracking; one confidence level for honest reporting.

6a. Composite Profile Score

The composite score is the unweighted mean of the three normalized subscale scores. An unweighted approach is used for principled reasons: the SECUR-E Developer Personas Framework and DASP research provide no empirical basis for a differential weighting scheme. The DASP study does not, for example, establish that OPP is systematically 1.5× more predictive of security behavior than MOT, and inventing weights without an empirical foundation would create false precision. The unweighted mean is transparent, reproducible, and honest about what the instrument can and cannot claim.

The composite score's function is limited and specific. It serves as a single-number summary for longitudinal tracking (comparing pre- and post-intervention profiles for the same individual or team) and for aggregate reporting across teams or organizations. It is never used as the basis for persona classification; that task requires a three-dimensional profile.

Subscale	Raw Score (8-40)	Normalized (0-100)	Band
Capability (CAP - Section B)	Enter score here	$= (\text{score} - 8) \div 32 \times 100$	(see band table)
Opportunity (OPP - Section C)	Enter score here	$= (\text{score} - 8) \div 32 \times 100$	(see band table)
Motivation (MOT - Section D)	Enter score here	$= (\text{score} - 8) \div 32 \times 100$	(see band table)
Composite Profile Score (unweighted mean)	Sum ÷ 3	Mean of three normalized scores	For aggregate reporting only

When the composite score is useful vs. misleading

Useful: Pre- and post-intervention comparison for the same individual. A +12-point composite gain after a targeted capability intervention on an Unaware profile is a meaningful signal.

Useful: Team-level aggregate. If a team's mean composite is 41 and rises to 54 after a six-month culture program, that is a plausible effect size worth reporting.

Misleading: Comparing two individuals with different profile shapes. A composite of 60 built from CAP=80, OPP=40, MOT=60 is a completely different situation from the same composite built from CAP=60, OPP=65, MOT=55. The first is an Enthusiast; the second is an ambiguous Compliant/Skeptic. Always report the profile alongside any composite.

6b. Confidence Rating

The confidence rating communicates how certain any given classification should be. It is a mandatory output of the scoring process, not an optional annotation. A classification without a confidence rating is incomplete because it suppresses critical information for intervention design. A low-confidence Skeptic classification should prompt a very different response from a practitioner than a High-confidence one.

Confidence	Criteria	Practitioner Action	Worked Example
High	All three subscale scores fall within the expected ranges for the primary persona. No single subscale score falls within the expected range of an adjacent persona. Applies to fewer than half of real-world profiles.	Report primary persona with high confidence. Proceed to intervention design (D6) for that persona.	CAP=34, OPP=36, MOT=35 → <i>Embedded. All three dimensions are unambiguously High. No plausible alternative.</i>
Moderate	Two of three subscale scores fall within the primary persona's expected ranges. One subscale is in a boundary zone (within ± 4 points of an adjacent persona's threshold). Applies to approximately half of the profiles.	Report primary persona with secondary persona risk flag. Consult the borderline table for the ambiguous dimension. Supplement with Reflection prompt responses.	CAP=31, OPP=28, MOT=33 → <i>Primary: Enthusiast. OPP is at the upper bound (threshold: ≤ 26). Flag Embedded risk if OPP items show high organizational support that is inconsistently leveraged.</i>
Low	Only one subscale falls cleanly within the primary persona's range, OR two personas produce equally plausible classifications, OR the profile does not match any single persona's pattern (atypical/mixed profile).	Report the two most plausible personas. Do not force a single classification. Use open reflection responses and Section A context to adjudicate. Consider a facilitated discussion if team deployment allows it.	CAP=25, OPP=25, MOT=24 → <i>Equidistant between Skeptic, Compliant, and Resistant. All moderate. Report as 'Undifferentiated moderate profile - Skeptic/Compliant most likely, Resistant risk present.'</i> <i>Prioritize Section E (Reflections - D2 Survey) for disambiguation.</i>

Worked Examples: Six Complete Profile Calculations

The table below walks through one hypothetical profile for each of the six SECUR-E personas, showing the raw scores, normalized scores, the specific rule or pattern that triggered the classification, the resulting persona and confidence level, and the intervention priority implied by the profile. These are not real participant data; they are constructed to illustrate how the scoring logic resolves in each canonical case. Real profiles will often be messier.

	Raw Scores	Norm. (0-100)	Rule Applied	Result - Confidence	Notes
Example A	CAP: 14 OPP: 19 MOT: 13	CAP: 19 OPP: 34 MOT: 16	Rule 1 (Priority 1): CAP $\leq$ 18 AND MOT $\leq$ 18.	Unaware High	Both CAP and MOT are Very Low. OPP is Low (not Very Low), suggesting some organizational infrastructure exists but cannot be productively engaged. Primary intervention: foundational capability building before any other lever (DASP D28, D33).
Example B	CAP: 33 OPP: 21 MOT: 36	CAP: 78 OPP: 41 MOT: 88	Rule 3: CAP $\geq$ 30 AND MOT $\geq$ 30 AND OPP $\leq$ 27.	Enthusiast High	Strong capability and very high motivation, but limited organizational opportunity. OPP of 21 confirms the Enthusiast pattern; the developer is acting against the grain of their environment. Burnout risk. Intervention priority: opportunity creation (DASP D3, D1).
Example C	CAP: 26 OPP: 34 MOT: 20	CAP: 56 OPP: 81 MOT: 38	Rule 4: OPP $\geq$ 29 AND MOT $\leq$ 24 AND CAP 22-29.	Compliant High	Moderate CAP, high OPP, low MOT. The organization is doing the motivational work. Security behavior is contingent on continued enforcement. Reflection 2 would likely reveal difficulty naming unprompted security initiatives. Intervention: shift from extrinsic to intrinsic motivation (DASP D8, D15, D16). Note: this profile now also falls within the widened Compliant→Embedded transitional trigger zone (Step 5: OPP $\geq$ 28, MOT 17-30, CAP 26-34). It stays plain Compliant here because Reflection 2 shows no unprompted initiative, a clean illustration that the quantitative trigger only flags candidates for the transitional check; the qualitative signal still decides it.
Example D	CAP: 27 OPP: 25 MOT: 14	CAP: 59 OPP: 53 MOT: 19	Rule 5: MOT $\leq$ 16 AND CAP 22-30 AND OPP 22-28.	Resistant Moderate	MOT at 14 is Very Low - check MOT-3/MOT-7 (items 19 and 23) for reactance signals. OPP of 25 is at the lower end of the Moderate range, which is expected. Note: If item 19 $\geq$ 4 (security overhead frustration), the Resistant classification is confirmed. Intervention: autonomy-preserving approaches (DASP D20, D21); coercive mandates will deepen resistance.
Example E	CAP: 25 OPP: 24 MOT: 21	CAP: 53 OPP: 50 MOT: 41	Rule 6: CAP 22-30, OPP 20-28, MOT 17-24. No higher-pri	Skeptic Moderate	All three dimensions are moderate; this is the hardest profile to read. MOT at 21 is Low-to-Moderate, placing it just above the Resistant boundary. Check MOT-7 (item 23: disproportionate security effort). A Skeptic will often agree moderately (3-4); a Resistant will agree strongly (5). Intervention: evidence-based persuasion (DASP D12, D13, D15).

	Raw Scores	Norm. (0-100)	Rule Applied	Result - Confidence	Notes
			riority rule triggered.		
Example F	CAP: 37 OPP: 35 MOT: 38	CAP: 91 OPP: 84 MOT: 94	Rule 2: CAP ≥ 30 AND OPP ≥ 28 AND MOT ≥ 30.	Embedded <i>High</i>	All three dimensions are High to Very High. Security is habitual, environmentally supported, and intrinsically valued. This developer is a candidate security champion. Intervention priority: sustain and amplify mentoring opportunities, formal champion roles (DASP D3, D30). Exemplar check (Rule 2a): CAP and MOT are Very High (≥37), but OPP=35 is High, not Very High; the Exemplar flag does not fire. Report as plain Embedded. A useful near-miss: raising OPP alone, not CAP or MOT, is what would tip this profile into Embedded (Exemplar).

Normalization values have been rounded to the nearest integer. In the composite score (Step 6a), compute the mean of the three exact normalized values before rounding to avoid accumulation of rounding error.

Important Interpretive Limits

Three interpretive limits should be explicitly acknowledged whenever this scoring system is applied in practice, whether self-administered or facilitated.

First, personas are heuristics, not labels. SECUR-E is explicit that 'personas should be understood as heuristic tools rather than rigid classifications.' A developer classified as Resistant based on a current score is not a Resistant person; they are a person whose current COM-B profile resembles the Resistant pattern under current conditions. Conditions change. Personas shift. For instance, a Compliant developer who is actively moving toward Embedded, and a developer who simultaneously exhibits Skeptic and Compliant characteristics depending on the organizational context. Never use a persona classification to make personnel decisions or to characterize a developer's general competence or attitude.

Second, self-report has known limitations. The instrument measures developers' perceptions of their own capabilities, opportunities, and motivations. DASP D32 explicitly flags overconfidence as a risk: 'extensive development experience can produce false confidence in the ability to effectively conduct a security practice without knowing the necessary security skill set required.' A developer scoring High on CAP through confident self-assessment may have real gaps that only behavioral observation would reveal. This does not make the self-report useless; perception shapes behavior as much as objective reality does, but it should inform how practitioners use results.

Third, the classification thresholds have not been empirically validated on a large developer population. To improve SECUR-E, as future work: 'quantitative survey research could assess the prevalence of personas and whether developers recognize themselves in persona descriptions.' The thresholds used in this system are grounded in the qualitative DASP interview data. However, they have not been tested for sensitivity and specificity with respect to behavioral outcomes, such as vulnerability rates or security tool adoption. Use them as a starting framework for diagnosis, not as an established clinical instrument.